

Automatic Verification of Differential Characteristics: Application to Reduced Gimli (Full Version)

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Abstract Since Keccak was selected as the SHA-3 standard, more and more permutation-based primitives have been proposed. Different from block ciphers, there is no round key in the underlying permutation for permutation-based primitives. Therefore, there is a higher risk for a differential characteristic of the underlying permutation to become incompatible when considering the dependency of difference transitions over different rounds. However, in most of the MILP or SAT based models to search for differential characteristics, only the difference transitions are involved and are treated as independent in different rounds, which may cause that an invalid one is found for the underlying permutation. To overcome this obstacle, we are motivated to design a model which automatically avoids the inconsistency in the search for differential characteristics. Our technique is to involve both the difference transitions and value transitions in the constructed model. Such an idea is inspired by the algorithm to find SHA-2 characteristics as proposed by Mendel et al. in ASIACRYPT 2011, where the differential characteristic and the conforming message pair are simultaneously searched. As a first attempt, our new technique will be applied to the Gimli permutation, which was proposed in CHES 2017. As a result, we reveal that some existing differential characteristics of reduced Gimli are indeed incompatible, one of which is found in the Gimli document. In addition, since only the permutation is analyzed in the Gimli document, we are lead to carry out a comprehensive study, covering the proposed hash scheme and the authenticated encryption (AE) scheme specified for Gimli, which has become a second round candidate of the NIST lightweight cryptography standardization process. For the hash scheme, a semi-free-start (SFS) collision attack can reach up to 8 rounds starting from an intermediate round. For the AE scheme, a state recovery attack is demonstrated to achieve up to 9 rounds. It should be emphasized that our analysis does not threaten the security of Gimli.

Keywords: Gimli, hash function, AE, MILP, collision, state-recovery

1 Introduction

As the demand for lightweight cryptographic primitives in industry increases, NIST is currently holding a public Lightweight Cryptography Standardization process [1], aiming at lightweight cryptography standardization by combining the efforts from both academia and industry. Among the 32 second round candidates, Gimli was first proposed in CHES 2017 [4]. The main strategy to improve its performance is to process the 384-bit data in four 96-bit columns independently and make only a 32-bit word swapping among the four columns every two rounds. Such a design strategy soon received a doubt from Hamburg [12]. However, the attack in [12] works for an ad-hoc mode rather than the proposed hash scheme or AE scheme in the submitted Gimli document.

Along the development of differential attacks [6], several variants have been proposed. A very influential one was the modular differential attack on the MD-SHA hash family, which directly turned MD5 [21] and SHA-1 [18,20] into broken hash functions. To mount collision attacks on MD5 and SHA-1 as in [20,21], one challenging work is to find a proper differential characteristic, which was first finished by hand-craft [20,21]. Later, the guess-and-determine method to search for differential characteristics was proposed in ASIACRYPT 2006, together with its application to full SHA-1 [9]. However, when such a guess-and-determine technique is directly applied to reduced SHA-2, Mendel et al. pointed out in [16] that the discovered differential characteristics are always invalid since contradictions may easily occur in the set of conditions implied in the discovered differential characteristics. To overcome this obstacle, they finally developed an algorithm to search for the differential characteristic and the conforming message pair simultaneously to avoid the inconsistency.

Indeed, such a case does not only exist in the MD-SHA hash family. For the ARX construction, for instance, some differential characteristics of Blake-256 [7] and Skein-512 [3] are also proven to be invalid if taking some dependency into account, as revealed by Leurent [14]. To search for valid differential characteristics of reduced Skein, Leurent designed a dedicated algorithm in [15] using the improved generalized conditions [14] and the guess-and-determine technique [9].

In another direction, since the introduction of the MILP-based method to search for differential characteristics [19], the SAT-based method has also been developed [13]. However, in most of the MILP models or SAT models to search for differential characteristics [4,13,19,23], only the difference transitions are taken into account and are treated as independent in different rounds. Although such an assumption is commonly believed to be reasonable for block ciphers, it may not hold well for permutation-based primitives since there is no round key in the permutation. A similar problem has been investigated in [8]. Moreover, since Keccak [5] was selected as the SHA-3 standard, more and more

permutation-based primitives have been proposed. However, whether similar cases once appearing in SHA-2 [16], Skein-512 [3] and Blake-256 [7] will occur in the commonly constructed MILP or SAT models to search for differential characteristics for the underlying permutation remains unknown. Therefore, it is vital to make an investigation for such a problem.

However, both the methods in [15,16] require a dedicated implementation of the heuristic search. In addition, how to achieve the simultaneousness is ambiguous in [16]. For [15], the inconsistency is avoided by using the improved generalized conditions [14]. As is known, the most convincing way is to provide a conforming message pair for the discovered differential characteristic.

Therefore, similar to the motivation to introduce the MILP-based method into cryptanalysis, it would be meaningful to utilize some off-the-shelf tools to reduce the workload. Consequently, we take Gimli as our first attempt and are motivated to tackle the problem of how to construct a model to **always** avoid the incompatibility in the search for differential characteristics. Moreover, since Gimli is one of the second round candidates in NIST Lightweight Cryptography Standardization process, we will provide some additional analysis of reduced Gimli. We noticed that there is a related work [17] for MD-SHA hash family published at SAT 2006 aiming at automatic message modification, though with ambiguous technical details.

Our Contributions. We made a comprehensive study of Gimli⁵, as summarized below:

- We make the first step to investigate the properties of the SP-box. Such a work is meaningful since all the attacks in this paper heavily rely on them.
- A novel MILP model capturing the difference transitions and value transitions simultaneously is developed. To the best of our knowledge, this is the first model which takes both transitions into account. This model can be simply used to detect contradictions in the differential characteristic of Gimli. As a result, we prove that both the 12-round differential characteristic in the Gimli document [4] and the 6-round differential characteristic used for the collision attack on 6-round Gimli-Hash in [23] are invalid. The second usage of this model is to directly search for a valid differential characteristic and the conforming message pair simultaneously.
- For the hash scheme, we provide the first practical semi-free-start (SFS) colliding message pair for 6-round Gimli-Hash and develop several techniques to convert SFS collisions into collisions. Moreover, we also mount a SFS collision attack on the intermediate 8-round Gimli-Hash.
- For the AE scheme, we are curious why the designers only claim 128-bit security while a 256-bit key is used. Thus, we are motivated to devise an attack which can maximize the number of rounds with complexity below 2^{256} . Consequently, we mount a state-recovery attack on 9-round Gimli with a rather high time complexity 2^{192} and memory complexity 2^{190} .

⁵ The source code of our attacks can be referred to https://github.com/LFKOK MI/Gimli_nalYSIS.git

The memory/data/time complexity of the above attacks are displayed in Table 1.

Organization. The Gimli permutation and some properties of the SP-box will be introduced in section 2 and section 3, respectively. Then, the MILP model capturing both difference transitions and value transitions will be described in section 4. The (SFS) collision attack on 6-round and 8-round Gimli-Hash will be shown in section 5 and section 6, respectively. Then, we will investigate the security of the AE scheme and present the state-recovery attack on 9-round Gimli in section 7. Finally, we conclude the paper in section 8.

Table 1: The analytical results of reduced Gimli, where Z-S represents Zero-sum and Z-D represents Zero-internal-difference.

Target	Attack Type	Rounds	Memory	Data	Time	Ref.
Hash scheme	SFS collision	6	practical			Sec. 5
Hash scheme	collision	6	2^{64}	-	2^{64}	Sec. 5.3
Hash scheme	collision	6	negligible	-	$2^{91.4}$	App. D
Hash scheme	SFS collision	8	negligible	-	2^{64}	Sec. 6
AE scheme	state-recovery	5	2^{126}	4	2^{128}	App. F
AE scheme	state-recovery	9	2^{19}	4	2^{192}	Sec. 7.1

2 Description of Gimli

The Gimli state can be viewed as a two-dimensional array $S = (S_{i,j})$ ($0 \leq i \leq 2, 0 \leq j \leq 3$), where $S_{i,j} \in F_2^{32}$, as illustrated in Figure 1. The Gimli permutation is specified with Algorithm 1 in Appendix A. As can be seen from the description of the Gimli permutation, the 24-round permutation can be viewed as iterating the following sequence of operations for 6 times:

$$(\text{SP} \rightarrow \text{S_SW} \rightarrow \text{AC}) \rightarrow (\text{SP}) \rightarrow (\text{SP} \rightarrow \text{B_SW}) \rightarrow (\text{SP}),$$

where the SP-box operation, Small-Swap operation, Big-Swap operation and AddRoundConstant operation are denoted by SP, S_SW, B_SW and AC, respectively. For convenience, denote the internal state after r -round permutation by S^r and the input state by S^0 . In other words, we have

$$S^{4i} \xrightarrow{\text{SP}} S^{4i+0.5} \xrightarrow{\text{S_SW}} \xrightarrow{\text{AC}} S^{4i+1} \xrightarrow{\text{SP}} S^{4i+2} \xrightarrow{\text{SP}} \xrightarrow{\text{S_BW}} S^{4i+3} \xrightarrow{\text{SP}} S^{4i+4},$$

where $0 \leq i \leq 5$. In addition, $\oplus$ denotes the exclusive or difference in S^r ($0 \leq r \leq 24$). $Z[i]$ ($0 \leq i \leq 31$) denotes the $(i+1)$ -th bit of the 32-bit word Z and $Z[0]$ is the least significant bit of Z . $Z[i \sim j]$ ($0 \leq j < i \leq 31$) represents the $(j+1)$ -th bit to the $(i+1)$ -th bit of the 32-bit word Z . For example, $Z[1 \sim 0]$ represents the two bits $(Z[1], Z[0])$. Moreover, $\oplus$, $\ll$, $\lll$, $\vee$ and $\wedge$ represent the logic operations *exclusive or*, *shift left*, *rotate left*, *or*, and *and*, respectively.

,	,1	,2	,3
1,	1,1	1,2	1,3
2,	2,1	2,2	2,3

Fig. 1: The Gimli state

2.1 SP-box

The SP-box of Gimli takes a 96-bit value as input and outputs a 96-bit value. Denote the input and the output by $(IX, IY, IZ) \in F_2^{32 \times 3}$ and $(OX, OY, OZ) \in F_2^{32 \times 3}$, respectively. Then, the relation between (OX, OY, OZ) and (IX, IY, IZ) can be described as follows:

$$\begin{aligned}
IX &\leftarrow IX \ll 24 \\
IY &\leftarrow IY \ll 9 \\
OZ &\leftarrow IX \oplus IZ \ll 1 \oplus (IY \wedge IZ) \ll 2 \\
OY &\leftarrow IY \oplus IX \oplus (IX \vee IZ) \ll 1 \\
OX &\leftarrow IZ \oplus IY \oplus (IX \wedge IY) \ll 3
\end{aligned}$$

Based on the above relation, the following bit relations can be derived, where the indices are considered within modulo 32.

$$\begin{aligned}
OX[i] = & \begin{aligned} & IZ[i] \oplus IY[i \ll 9] \quad (0 \leq i \leq 2) \\ & IZ[i] \oplus IY[i \ll 9] \oplus (IX[i \ll 27] \wedge IY[i \ll 12]) \quad (3 \leq i \leq 31) \end{aligned} \quad (1)
\end{aligned}$$

$$\begin{aligned}
OY[i] = & \begin{aligned} & IY[i \ll 9] \oplus IX[i \ll 24] \quad (i = 0) \\ & IY[i \ll 9] \oplus IX[i \ll 24] \oplus (IX[i \ll 25] \vee IZ[i \ll 1]) \quad (1 \leq i \leq 31) \end{aligned} \quad (2)
\end{aligned}$$

$$\begin{aligned}
OZ[i] = & \begin{cases} IX[i \ll 24] & (i = 0) \\ IX[i \ll 24] \oplus IZ[i \ll 1] & (i = 1) \\ IX[i \ll 24] \oplus IZ[i \ll 1] \oplus (IY[i \ll 11] \wedge IZ[i \ll 2]) & (2 \leq i \leq 31) \end{cases} \quad (3)
\end{aligned}$$

2.2 Linear Layer

The linear layer includes two different swap operations, namely Small-Swap and Big-Swap. Small-Swap occurs every 4 rounds starting from the 1st round. Big-Swap occurs every 4 rounds starting from the 3rd round. The illustration of Small-Swap and Big-Swap can be referred to Figure 2.

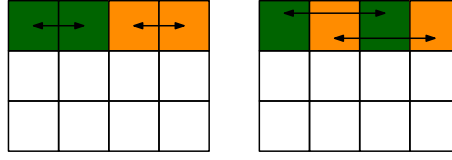


Fig. 2: The linear layer. The left/right one represent the Small-Swap/Big-Swap.

2.3 Gimli-Hash

How Gimli-Hash compresses a message is illustrated in Figure 3. Specifically, Gimli-Hash initializes a 48-byte Gimli state to all-zero. It then reads sequentially through a variable-length input as a series of 16-byte input blocks, denoted by $M_0, M_1, \dots$. After all message blocks are processed, the 256-bit hash value will be generated. More details can be referred to [1].

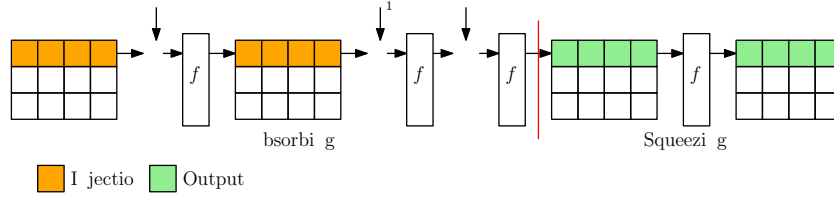


Fig. 3: The process to compress the message, where f is the Gimli permutation

3 Properties of the SP-box

Since several properties of the SP-box will be exploited in our collision attack and state-recovery attack, for convenience, we summarize them in this part. For simplicity, the input and output of the SP-box are denoted by (IX, IY, IZ) and (OX, OY, OZ) , respectively.

Property 1 *If $IY[31 \sim 23] = 0$ and $IY[19 \sim 0] = 0$, OX will be independent of IX .*

Property 2 *A random triple (IY, IZ, OX) is potentially valid with probability $2^{-15.5}$ without knowing IX .*

Property 3 *Given a random triple (IX, OY, OZ) , it is valid with probability 2^{-1} . Once it is valid, $(OX[30 \sim 0], IY, IZ[30 \sim 0])$ can be determined.*

Property 4 *Given a random triple (IY, IZ, OZ) , (IX, OX, OY) can be uniquely determined. In addition, a random tuple (IY, IZ, OY, OZ) is valid with probability 2^{-32} .*

Property 5 Suppose the pair (IY, IZ) and t bits of OY are known. Then t bits of information on IX can be recovered by solving a linear equation system of size t .

The above properties will be frequently exploited in our attacks and therefore we list them ahead of time. The corresponding proofs can be referred to Appendix B. Some other properties will be explained later.

4 The MILP Model Capturing Difference and Value Transitions

To search for a valid differential characteristic of reduced SHA-2, Mendel et al. developed a technique to search for the differential characteristic and conforming message pair simultaneously [16]. However, how to achieve the simultaneousness is not explained in [16]. Inspired by such an idea, different from many models where only the difference transitions are considered and are treated as independent in different rounds, we try to construct a model which can describe the difference transitions and value transitions simultaneously. The basic idea is simple. As shown in Figure 4, the models to describe the difference transitions and value transitions will be independently constructed. Then, construct a model to describe the difference-value relations in the nonlinear operation and use it to connect the difference transitions and value transitions. The reason is that the difference transitions and value transitions are dependent only in the nonlinear operation. If such a model can be constructed, the contradictions can always be avoided in the search.

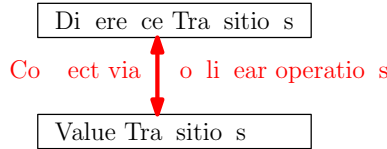


Fig. 4: Illustration of the model

4.1 Difference-Value Relations Through the SP-box

First of all, consider the relations between the difference and value. According to the bit relations between (IX, IY, IZ) and (OX, OY, OZ) as specified in Equation 1, Equation 2, and Equation 3, one can easily observe that there are at most 4 types of Boolean expressions as follows, where $a[i] \in F_2$ and $0 \leq i \leq 4$.

Type-1: $a[1] = a[0]$.

Type-2: $a[2] = a[0] \oplus a[1]$.